

Problem 1.

Solution:

- (a) Two battlefields with values $(v_1, v_2) = (3, 4)$. Player 1 (row player) has budget $b_1 = 3$; Player 2 (column player) has budget $b_2 = 4$. A pure strategy is an integer split of the budget across the two battlefields.

- P1 rows $R = \{(0, 3), (1, 2), (2, 1), (3, 0)\}$
- P2 columns $C = \{(0, 4), (1, 3), (2, 2), (3, 1), (4, 0)\}$

Let $A \in \mathbb{R}^{4 \times 5}$ be player 2's payoff matrix, whose (r, c) entry is $\sum_{i=1}^2 v_i \cdot \text{sgn}(y_i - x_i)$ for row $r = (x_1, x_2)$ and column $c = (y_1, y_2)$. For this instance

	(0, 4)	(1, 3)	(2, 2)	(3, 1)	(4, 0)
(0, 3)	4	3	-1	-1	-1
(1, 2)	1	4	3	-1	-1
(2, 1)	1	1	4	3	-1
(3, 0)	1	1	1	4	3

LPs for the optimal mixed strategies and the computed solution

Column player (Player 2) LP (primal).

$$\begin{aligned} \max_{p \in \mathbb{R}^5, v \in \mathbb{R}} \quad & v \\ \text{s.t.} \quad & A p \geq v \mathbf{1}_4, \\ & \mathbf{1}_5^\top p = 1, \quad p \geq 0. \end{aligned}$$

Row player (Player 1) LP (dual).

$$\begin{aligned} \min_{q \in \mathbb{R}^4, w \in \mathbb{R}} \quad & w \\ \text{s.t.} \quad & q^\top A \leq w \mathbf{1}_5^\top, \\ & \mathbf{1}_4^\top q = 1, \quad q \geq 0. \end{aligned}$$

Running gurobi in python using the parameters above gives the results:

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1  ----Player 1----
2  Optimal value w* = 1.81
3  q[(0,3)] = 0.27
4  q[(1,2)] = 0.09
5  q[(2,1)] = 0.39
6  q[(3,0)] = 0.25
7
8  ----Player 2----
9  Optimal value v* = 1.81
10 p[(0,4)] = 0.25
11 p[(1,3)] = 0.39
12 p[(2,2)] = 0.09
13 p[(3,1)] = 0.27
14 p[(4,0)] = 0.00

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Payoff to Player 1: $-v^* = -u^* = -1.81$, Payoff to Player 2: $v^* = u^* = 1.81$.

- (b) General k -battlefield game with integer budgets b_1, b_2 . Let the battlefield values be $\mathbf{v} = (v_1, \dots, v_k)$ with $v_i \in \mathbb{Z}_{\geq 1}$. Define the sets of pure strategies for the two players:

$$S_1 = \left\{ s = (s_1, \dots, s_k) \in \mathbb{Z}_{\geq 0}^k : \sum_{i=1}^k s_i = b_1 \right\}, \quad S_2 = \left\{ t = (t_1, \dots, t_k) \in \mathbb{Z}_{\geq 0}^k : \sum_{i=1}^k t_i = b_2 \right\}.$$

For a row $s \in S_1$ and a column $t \in S_2$, define the payoff on each battlefield $r \in \{1, \dots, k\}$ by comparing allocations s_r (P_1) and t_r (P_2):

$$\text{payoff to } P_2 = \begin{cases} +v_r, & t_r > s_r, \\ -v_r, & t_r < s_r, \\ 0, & t_r = s_r. \end{cases}$$

The total payoff to P_2 is the sum across battlefields:

$$A_{s,t} = \sum_{r=1}^k v_r \text{sgn}(t_r - s_r), \quad \text{where} \quad \text{sgn}(u) = \begin{cases} 1, & u > 0, \\ 0, & u = 0, \\ -1, & u < 0. \end{cases}$$

Therefore $A \in \mathbb{R}^{|S_1| \times |S_2|}$ is player 2's payoff matrix.

Player 2 (column player)

$$\begin{aligned} \max_{x \in \mathbb{R}^{|S_2|}, v \in \mathbb{R}} \quad & v \\ \text{s.t.} \quad & Ax \geq v \mathbf{1}_{|S_1|}, \\ & \mathbf{1}_{|S_2|}^\top x = 1, \quad x \geq 0. \end{aligned}$$

Player 1 (row player)

$$\begin{aligned} \min_{y \in \mathbb{R}^{|S_1|}, u \in \mathbb{R}} \quad & u \\ \text{s.t.} \quad & y^\top A \leq u \mathbf{1}_{|S_2|}^\top, \\ & \mathbf{1}_{|S_1|}^\top y = 1, \quad y \geq 0. \end{aligned}$$

By the minimax theorem,

$\max_x \min_i (Ax)_i = \min_y \max_j (y^\top A)_j = \text{Value of the game}$
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Problem 2.

Solution: Let A be the column player's (cp) payoff matrix:

$$A = \begin{array}{c|ccc} & P & S & R \\ \hline P & 0 & -1 & 1 \\ S & 1 & 0 & -1 \\ R & -1 & 1 & 0 \end{array}.$$

A mixed strategy for cp is $x = (x_P, x_S, x_R)^\top$, and for the row player (rp) is $y = (y_P, y_S, y_R)^\top$. The cp payoff under (y, x) is $y^\top Ax$.

$$\begin{array}{ll} \text{cp: } \max_{x,v} & v \\ \text{s.t.} & [Ax]_P \geq v, \\ & [Ax]_S \geq v, \\ & [Ax]_R \geq v, \\ & x_P + x_S + x_R = 1, \\ & x_P, x_S, x_R \geq 0; \end{array} \quad \begin{array}{ll} \text{rp: } \min_{y,u} & u \\ \text{s.t.} & [y^\top A]_P \leq u, \\ & [y^\top A]_S \leq u, \\ & [y^\top A]_R \leq u, \\ & y_P + y_S + y_R = 1, \\ & y_P, y_S, y_R \geq 0. \end{array}$$

Let

$$z = \begin{bmatrix} x \\ v \end{bmatrix}, \quad w = \begin{bmatrix} y \\ u \end{bmatrix}.$$

Then the problems can be compactly written as

$$\begin{array}{ll} \max[1, 0]z & \min[1, 0]w \\ \text{s.t. } (1, -A)z \leq 0, & \text{s.t. } (1, -A^\top)w \geq 0, \\ (0, 1^\top)z = 1, & (0, 1^\top)w = 1, \\ x \geq 0; & y \geq 0. \end{array} \quad \begin{array}{l} \text{(cp)} \\ \text{(rp')} \end{array}$$

Dual of (cp').

$$\begin{array}{ll} \min 0^\top \lambda + 1 \cdot \theta = \theta \\ \text{s.t.} & 1^\top \lambda = 1, \\ & -A^\top \lambda + \theta \cdot 1 \geq 0, \\ & \lambda \geq 0, \theta \text{ free.} \end{array}$$

Rename $y = \lambda$, $u = \theta$. Then the dual becomes

$$\min_{y,u} u \quad \text{s.t.} \quad y^\top A \leq u 1^\top, \quad 1^\top y = 1, \quad y \geq 0,$$

which the same as the row player problem (rp').

Dual of (rp').

$$\begin{aligned} \max \quad & 0^\top \lambda - 1 \cdot \theta = -\theta \\ \text{s.t.} \quad & 1^\top \lambda = 1, \\ & A\lambda + \theta \cdot 1 \geq 0, \\ & \lambda \geq 0, \theta \text{ free.} \end{aligned}$$

Rename $x = \lambda$, $v = -\theta$. Then the dual becomes

$$\max_{x,v} v \quad \text{s.t.} \quad Ax \geq v \cdot 1, \quad 1^\top x = 1, \quad x \geq 0,$$

which is the same as the column player problem (cp').

Duality relationships.

$$\begin{aligned} \text{dual}(\text{cp}) &= \text{rp}, & \text{dual}(\text{rp}) &= \text{cp}. \\ \text{Opt}(\text{dual}(\text{cp})) &= \text{Opt}(\text{rp}), & \text{Opt}(\text{dual}(\text{rp})) &= \text{Opt}(\text{cp}). \end{aligned}$$

Weak Duality Theorem.

$$\begin{aligned} \text{Opt}(\text{cp}) &\leq \text{Opt}(\text{rp}), & \text{Opt}(\text{rp}) &\geq \text{Opt}(\text{cp}). \\ \text{Opt}(\text{cp}) &= \text{Opt}(\text{rp}). \end{aligned}$$

Strong Duality Theorem. Since cp and rp are feasible and bounded,

$$\text{Opt}(\text{cp}) = \text{Opt}(\text{dual}(\text{cp})), \quad \text{Opt}(\text{rp}) = \text{Opt}(\text{dual}(\text{rp})).$$

Therefore,

$$\boxed{\text{Opt}(\text{cp}) = \text{Opt}(\text{rp})}$$

Problem 3.

Solution:

(a) Primal:

$$\begin{aligned} \max \quad & 3x_1 + 2x_2 + x_3 \\ \text{s.t.} \quad & x_1 + 2x_2 + x_3 \leq 4, \\ & 2x_1 + x_2 + 3x_3 \leq 6, \\ & x_1, x_2, x_3 \geq 0. \end{aligned}$$

Dual:

$$\min \quad 4y_1 + 6y_2 \quad \text{s.t.} \quad \begin{cases} y_1 + 2y_2 \geq 3, \\ 2y_1 + y_2 \geq 2, \\ y_1 + 3y_2 \geq 1. \end{cases}$$

(1) $x^* = (\frac{8}{3}, \frac{2}{3}, 0)$

Both primal constraints are tight:

$$x_1 + 2x_2 + x_3 = 4, \quad 2x_1 + x_2 + 3x_3 = 6.$$

By complementary slackness, for $x_1, x_2 > 0$ we must have

$$y_1 + 2y_2 = 3, \quad 2y_1 + y_2 = 2 \implies y_1 = \frac{1}{3}, y_2 = \frac{4}{3}.$$

Check the remaining dual inequality:

$$y_1 + 3y_2 = \frac{13}{3} \geq 1$$

Objective values match:

$$3x_1 + 2x_2 + x_3 = \frac{28}{3} \quad \text{and} \quad 4y_1 + 6y_2 = \frac{28}{3}.$$

$x^* = (\frac{8}{3}, \frac{2}{3}, 0) \text{ is optimal, with } y^* = (\frac{1}{3}, \frac{4}{3})$

(2) $x^* = (3, 0, 0)$

Primal feasibility and tightness:

$$x_1 + 2x_2 + x_3 = 3 < 4 \quad (\text{slack}), \quad 2x_1 + x_2 + 3x_3 = 6 \quad (\text{tight}).$$

Since $x_1 > 0$, $x_2 = 0$, $x_3 = 0$, we apply complementary slackness conditions. For positive x_1 , the dual constraint corresponding to x_1 must hold with equality:

$$y_1 + 2y_2 = 3.$$

Because the first primal constraint is slack, its dual variable must be zero:

$$y_1 = 0 \implies y_2 = \frac{3}{2}.$$

Check dual feasibility for the remaining inequalities:

$$2y_1 + y_2 = \frac{3}{2} \not\geq 2, \quad y_1 + 3y_2 = \frac{9}{2} \geq 1.$$

The first violates dual feasibility. Therefore, no dual-feasible y satisfies the CS system for $x^* = (3, 0, 0)$.

$x^* = (3, 0, 0) \text{ is not optimal.}$

(b) Primal with free variables:

$$\begin{aligned} \max \quad & 2x_1 + x_2 + x_3 \\ \text{s.t.} \quad & x_1 + x_3 \leq 3, \\ & x_2 - x_3 \leq 2, \\ & x_1, x_2, x_3 \in \mathbb{R} \text{ (free)}. \end{aligned}$$

Dual:

$$\min 3y_1 + 2y_2 \quad \text{s.t.} \quad \begin{cases} y_1 = 2, \\ y_2 = 1, \\ y_1 - y_2 = 1. \end{cases}$$

Therefore the unique dual feasible solution is $(y_1, y_2) = (2, 1)$ with value 8, the primal optimum value is 8. Since $y_1, y_2 > 0$, both primal constraints must be tight at any primal optimum:

$$x_1 + x_3 = 3, \quad x_2 - x_3 = 2.$$

Along this line, the objective is

$$2x_1 + x_2 + x_3 = 2(3 - x_3) + (2 + x_3) + x_3 = 8,$$

so every point satisfying the two equalities is optimal.

(1) $x^* = (2, 3, 1)$

Primal feasibility and tightness:

$$x_1 + x_3 = 2 + 1 = 3 = 3 \quad (\text{tight}), \quad x_2 - x_3 = 3 - 1 = 2 = 2 \quad (\text{tight}).$$

Both constraints are tight and $y_1, y_2 > 0$. By complementary slackness, since both constraints are tight, there is no violation, the CS system holds perfectly.

Dual feasibility:

$$\begin{aligned} y_1 &= 2, & y_2 &= 1, \\ 3y_1 + 2y_2 &= 8, & \text{dual objective matches primal objective } (2x_1 + x_2 + x_3 &= 8). \end{aligned}$$

$x^* = (2, 3, 1) \text{ is optimal, with } y^* = (2, 1)$

(2) $\tilde{x} = (0, 0, 0)$

Primal feasibility:

$$x_1 + x_3 = 0 < 3 \quad (\text{slack}), \quad x_2 - x_3 = 0 < 2 \quad (\text{slack}).$$

Both constraints are slack. By complementary slackness, this requires $y_1 = y_2 = 0$.

However, with $y_1 = y_2 = 0$, the dual constraints

$$\begin{aligned} y_1 &= 2, \\ y_2 &= 1, \\ y_1 - y_2 &= 1 \end{aligned}$$

cannot be satisfied. Therefore no dual-feasible y satisfies the CS system.

$\tilde{x} = (0, 0, 0) \text{ is not optimal.}$

Solution:

- (a) Model a market with n buyers and m sellers. Each buyer i has valuation v_{ij} for a unit sold by seller j . Let $x_{ij} \in [0, 1]$ denote the allocation variable indicating whether buyer i purchases from seller j .

Primal:

$$\begin{aligned} \max_{x \geq 0} \quad & \sum_{i=1}^n \sum_{j=1}^m v_{ij} x_{ij} \\ \text{s.t.} \quad & \sum_{j=1}^m x_{ij} \leq 1 \quad \forall i = 1, \dots, n, \\ & \sum_{i=1}^n x_{ij} \leq 1 \quad \forall j = 1, \dots, m. \end{aligned}$$

Dual:

$$\begin{aligned} \min_{u, p \geq 0} \quad & \sum_{i=1}^n u_i + \sum_{j=1}^m p_j \\ \text{s.t.} \quad & u_i + p_j \geq v_{ij}, \quad \forall i, j. \end{aligned}$$

At equilibrium, (x, u, p) satisfy complementary slackness:

$$x_{ij} > 0 \Rightarrow u_i + p_j = v_{ij}, \quad x_{ij} = 0 \Rightarrow u_i + p_j > v_{ij}.$$

Results:

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1  ----Player 2----
2  Iteration 1: value = 99.00, M = 10
3  Iteration 2: value = 99.00, M = 10
4  Iteration 3: value = 98.00, M = 10
5  Iteration 4: value = 99.00, M = 10
6  Iteration 5: value = 99.00, M = 10
7  Iteration 6: value = 99.00, M = 10
8  Iteration 7: value = 99.00, M = 10
9  Iteration 8: value = 97.00, M = 10
10 Iteration 9: value = 99.00, M = 10
11 Iteration 10: value = 99.00, M = 10
12 Iteration 11: value = 97.00, M = 10
13 Iteration 12: value = 99.00, M = 10
14 Iteration 13: value = 97.00, M = 10
15 Iteration 14: value = 98.00, M = 10
16 Iteration 15: value = 99.00, M = 10
17 Iteration 16: value = 99.00, M = 10
18 Iteration 17: value = 95.00, M = 10
19 Iteration 18: value = 96.00, M = 10
20 Iteration 19: value = 99.00, M = 10
21 Iteration 20: value = 98.00, M = 10
22 Iteration 21: value = 99.00, M = 10
23 Iteration 22: value = 100.00, M = 10
24 Iteration 23: value = 98.00, M = 10

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25 Iteration 24: value = 96.00, M = 10
26 Iteration 25: value = 99.00, M = 10
27 Iteration 26: value = 98.00, M = 10
28 Iteration 27: value = 99.00, M = 10
29 Iteration 28: value = 99.00, M = 10
30 Iteration 29: value = 99.00, M = 10
31 Iteration 30: value = 98.00, M = 10
32 Iteration 31: value = 99.00, M = 10
33 Iteration 32: value = 100.00, M = 10
34 Iteration 33: value = 98.00, M = 10
35 Iteration 34: value = 100.00, M = 10
36 Iteration 35: value = 99.00, M = 10
37 Iteration 36: value = 99.00, M = 10
38 Iteration 37: value = 97.00, M = 10
39 Iteration 38: value = 99.00, M = 10
40 Iteration 39: value = 97.00, M = 10
41 Iteration 40: value = 98.00, M = 10
42 Iteration 41: value = 98.00, M = 10
43 Iteration 42: value = 97.00, M = 10
44 Iteration 43: value = 99.00, M = 10
45 Iteration 44: value = 99.00, M = 10
46 Iteration 45: value = 97.00, M = 10
47 Iteration 46: value = 98.00, M = 10
48 Iteration 47: value = 98.00, M = 10
49 Iteration 48: value = 99.00, M = 10
50 Iteration 49: value = 99.00, M = 10
51 Iteration 50: value = 98.00, M = 10

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- (b) Given a valuation matrix $V \in \mathbb{R}^{n \times m}$, let (x^*, u^*, p^*) denote one equilibrium obtained from the primal dual pair of the assignment game:

Primal:

$$\begin{aligned}
 \max_{x \geq 0} \quad & \sum_{i=1}^n \sum_{j=1}^m v_{ij} x_{ij} \\
 \text{s.t.} \quad & \sum_{j=1}^m x_{ij} \leq 1 \quad \forall i, \quad \sum_{i=1}^n x_{ij} \leq 1 \quad \forall j,
 \end{aligned}$$

Dual:

$$\begin{aligned}
 \min_{u, p \geq 0} \quad & \sum_{i=1}^n u_i + \sum_{j=1}^m p_j \\
 \text{s.t.} \quad & u_i + p_j \geq v_{ij} \quad \forall i, j.
 \end{aligned}$$

We wish to see whether the same instance admits another equilibrium different from (x^*, u^*, p^*) . The procedure consists of two tests.

Step 1: Search for another optimal matching.

- (a) Solve the primal LP once to obtain the optimal welfare value v^* and the set of matched pairs

$$M^* = \{(i, j) \mid x_{ij}^* > 0\}.$$

- (b) Re-solve the primal while explicitly excluding this matching via the additional constraint

$$\sum_{(i,j) \in M^*} x_{ij} \leq |M^*| - 1.$$

- (c) If the new optimum v' satisfies $|v' - v^*| \leq 10^{-6}$, then there exists another allocation x' that attains the same welfare. This implies the existence of a **different equilibrium with another matching**.

Step 2: Search for another equilibrium price vector.

- (a) Solve the dual again but fix the objective to the known optimum:

$$\sum_i u_i + \sum_j p_j = v^*.$$

- (b) To move along the face of optimal dual solutions, maximize a random linear functional

$$\max \sum_i r_i u_i + \sum_j s_j p_j$$

subject to the same feasibility constraints $u_i + p_j \geq v_{ij}$ and nonnegativity.

- (c) Let $(\tilde{u}, \tilde{p})$ be the new dual optimum. If

$$\|\tilde{u} - u^*\|_1 + \|\tilde{p} - p^*\|_1 > 10^{-6},$$

then another dual-optimal price vector exists, yielding a **different equilibrium with the same matching**.

Results:

1	Another equilibrium exists: True
2	Type: different prices (L1 distance=100)

- (c) Let (M^*, p^*) be a competitive equilibrium for an instance I of the assignment game. Let u_i^* denote the utility of buyer i under prices p^* . Then (u^*, p^*) is an optimal solution to the dual of the assignment linear program and satisfies

$$u_i^* + p_j^* \geq v_{ij} \quad \text{for all } (i, j), \quad u_i^* + p_j^* = v_{ij} \quad \text{for every } (i, j) \in M^*.$$

We call an edge *tight* if $u_i^* + p_j^* = v_{ij}$.

For any matching M , we have

$$\sum_{(i,j) \in M} v_{ij} \leq \sum_{(i,j) \in M} (u_i^* + p_j^*) \leq \sum_i u_i^* + \sum_j p_j^*.$$

Therefore, the weight of any matching is bounded above by the dual value $\sum_i u_i^* + \sum_j p_j^*$. Since M^* attains this bound, the dual value equals the maximum weight of a matching.

If we let $\overline{M}$ be any maximum-weight matching. Because $\overline{M}$ has the same total weight as M^* , equality must hold in the above inequality for every edge in $\overline{M}$. Thus, every edge $(i, j) \in \overline{M}$ is tight:

$$v_{ij} = u_i^* + p_j^*.$$

Tightness means that at prices p^* , every buyer in $\overline{M}$ is matched to one of their most preferred houses, and every seller receives the corresponding equilibrium price. Therefore, no buyer-seller pair can profitably deviate.

Hence, for any maximum-weight matching $\overline{M}$, $(\overline{M}, p^*)$ is also an equilibrium for I .